EXPERIMENT 06

Automating Business Processes Using Deluge Scripting and Workflows, with Application Testing and Debugging

Patient Appointment Booking Form — Workflow Automation Use Case

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1. Aim

To automate a key business process of the Patient Appointment & Scheduling Management application — sending an appointment confirmation email the moment a patient submits the Patient Profile form — using a Zoho Creator workflow and a Deluge script, and to test and debug the automation end-to-end with real form submissions.

This removes the need for front-desk staff to manually notify every patient, ensuring each submitted appointment request is acknowledged immediately and consistently.

2. Procedure

Step 1: Start a new workflow — From the application's Workflow section, a new workflow was created on the Patient Profile form.

Step 2: Define the trigger — The Record Event was set to Created or Edited and the Form Event (When to trigger workflow) was set to Successful form submission, so the automation fires only after a valid, saved submission.

Step 3: Name the workflow — The workflow was named Send Appointment Confirmation Email and saved using Create Workflow.

Step 4: Add a new action — In the Add New Action dialog, Run this action was left on Always, and Deluge script was chosen as the Action type so custom logic could compose and send the email.

Step 5: Write the Deluge script — A sendmail script was written in the Deluge editor: it sets from to zoho.loginuserid, to to input.email_address, builds a Subject of "Appointment Booking Confirmation – Patient Appointment & Scheduling Management", and concatenates a message with the patient's name, age, gender, department, preferred date and time, existing conditions, and symptoms pulled directly from the form's input fields.

Step 6: Save and review the action — The script was saved with Update, and the workflow summary confirmed the single Deluge script action was attached under Patient Profile › Created › Successful form submission › Send Appointment Confirmation Email, set to Execute without condition.

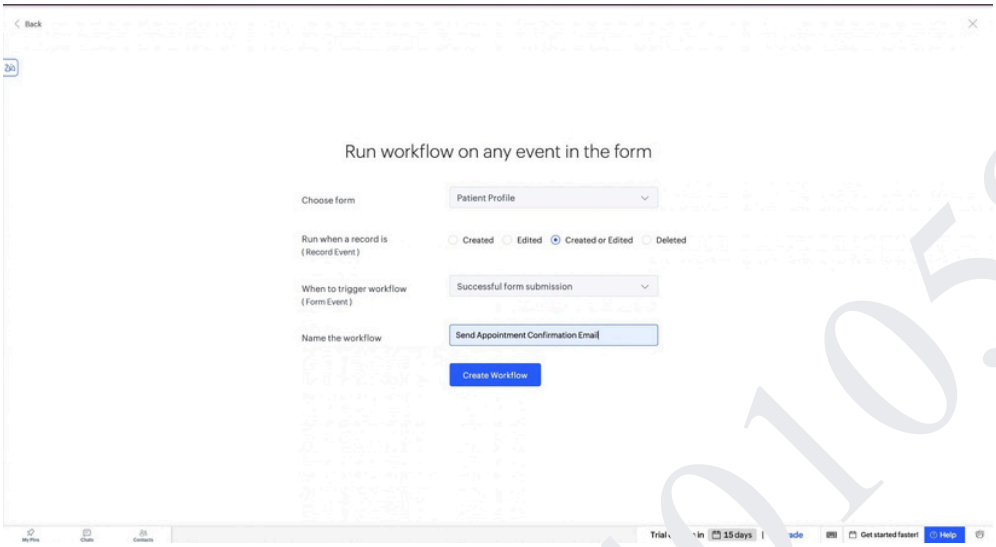
Step 7: Test with a live submission — A new Patient Profile record was filled in through the live form — Department: Dermatology, Existing Conditions: Allergy to Penicillin, Diabetes, Heart Disease, Asthma, Preferred Appointment Date: 04-Oct-2026, Consent For Treatment checked — and submitted to trigger the workflow.

Step 8: Debug and verify the output — The patient's inbox was checked immediately after submission to confirm the automation ran correctly: the confirmation email arrived with the correct dynamic values (name, age, gender, department, date, time, and conditions), confirming the Deluge script executed without errors.

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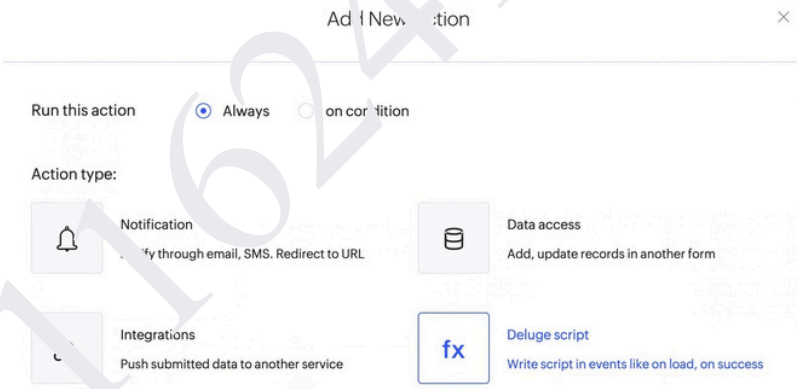
3. Output

3.1 Creating the workflow — trigger configuration



The new-workflow screen shows the form set to Patient Profile, the Record Event set to Created or Edited, and the Form Event set to Successful form submission, with the workflow named Send Appointment Confirmation Email. This defines exactly when the automation should run.

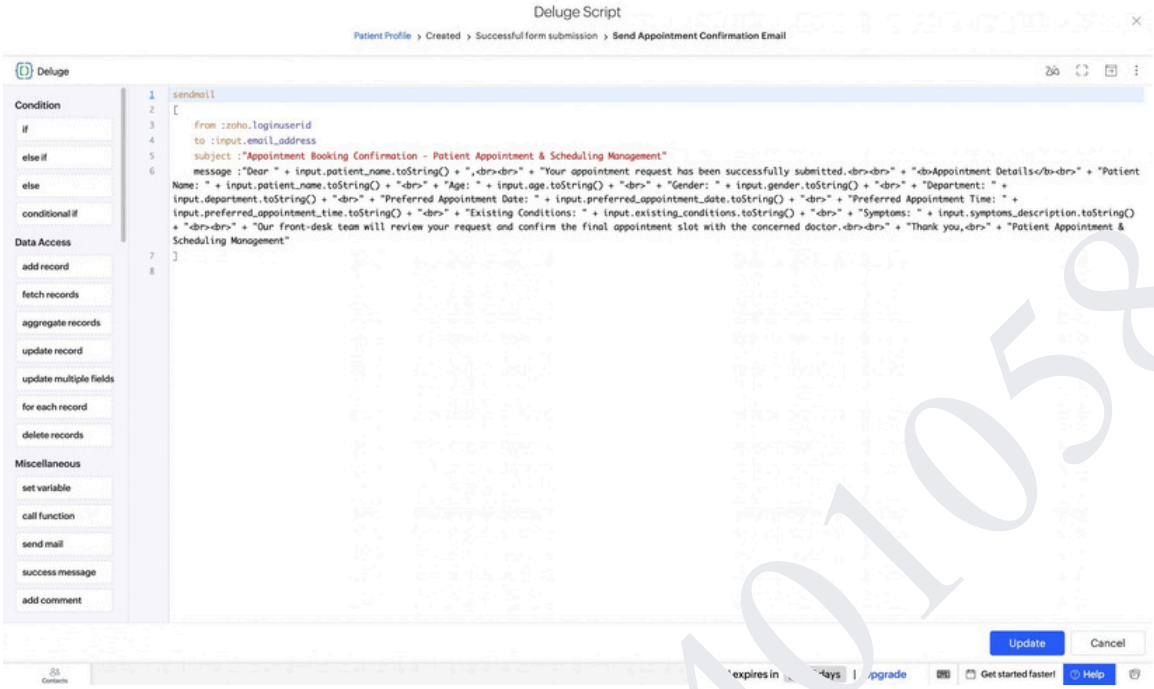
3.2 Choosing the action type



The Add New Action dialog lists the four available action types — Notification, Data access, Integrations, and Deluge script. Deluge script is selected here because the confirmation message needs dynamic, field-driven content that a simple notification template cannot build.

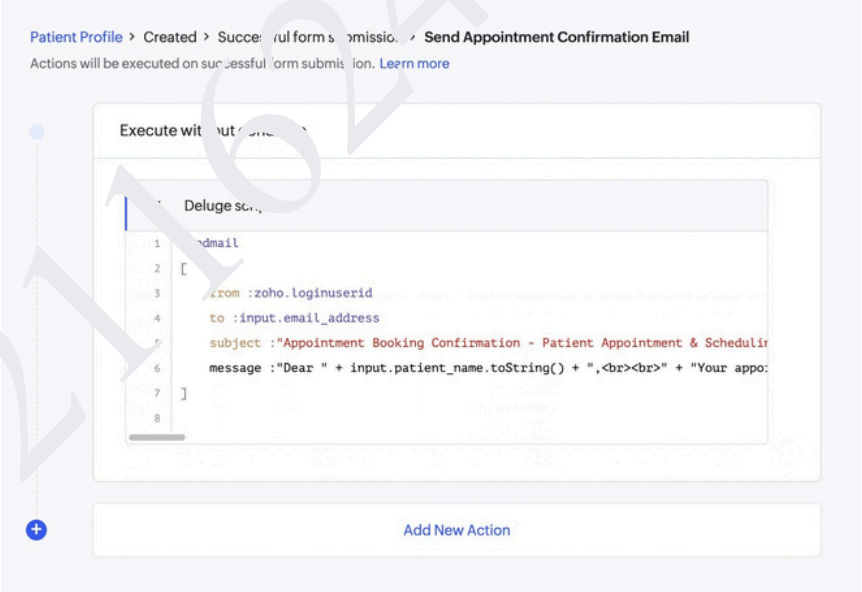
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3.3 Writing the Deluge script



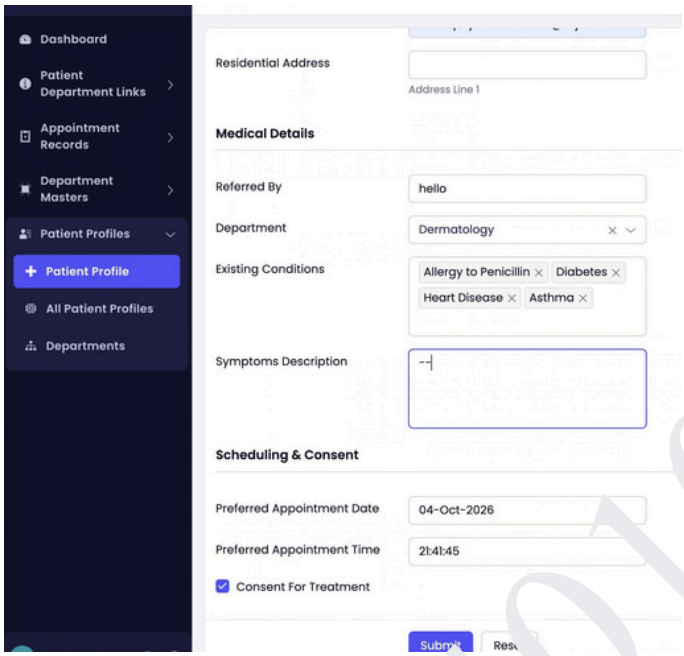
The Deluge editor shows the completed sendmail script, with the Condition, Data Access, and Miscellaneous action palette on the left for reference. The script reads input.email_address, input.patient_name, input.age, and the other form fields to build the subject and message dynamically.

3.4 Saved workflow action — summary view



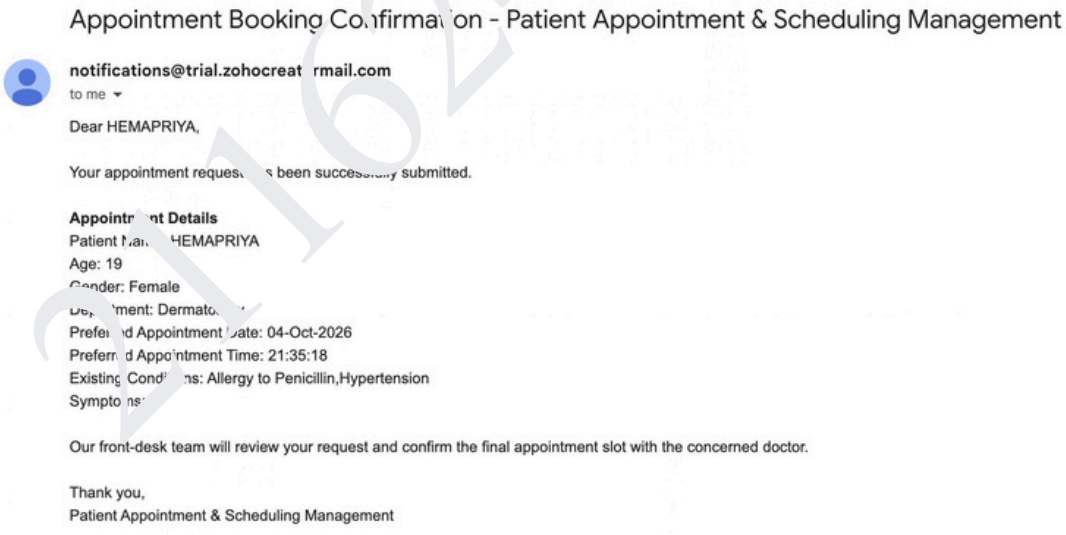
After saving, the workflow breadcrumb confirms the full path: Patient Profile › Created › Successful form submission › Send Appointment Confirmation Email, executing without condition. This view is also where a new action could be chained in if more automation were needed.

3.5 Testing — submitting a live record



A real Patient Profile record was filled in and submitted through the live form (Department: Dermatology, four Existing Conditions selected, Preferred Appointment Date 04-Oct-2026, Consent checked) specifically to trigger the Send Appointment Confirmation Email workflow and observe its behaviour.

3.6 Debugging output — confirmation email received



The received email confirms the automation ran successfully: the subject and message match the Deluge script exactly, and every dynamic value (Patient Name: HEMAPRIYA, Age: 19, Gender: Female, Department: Dermatology, date, time, and existing conditions) was pulled correctly from the submitted record, verifying the workflow end-to-end.

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4. Final Result

A complete workflow was built on the Patient Profile form that fires automatically on successful submission, using a Deluge sendmail script to generate a personalised confirmation email with no manual intervention from the front desk.

The automation was tested with a live form submission and debugged by inspecting the resulting email: all dynamic fields rendered correctly, confirming the script's field references and string concatenation were error-free.

The exercise met its objective: a real business process — notifying patients that their appointment request was received — is now fully automated through low-code workflow and Deluge scripting, and the testing step confirmed the automation behaves reliably end-to-end.

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